

SATYA VENKATESH KOSARAJU

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SUMMARY

Quality Technician at Danfoss Power Solutions (contract via Aerotek) with an M.S. in Agricultural & Biosystems Engineering and 3 years of hands-on test engineering experience - designing and instrumenting test rigs, writing and executing test plans, and documenting results in engineering reports for industry partners. Strong in fluid and granular flow modeling (DEM, CFD fundamentals), DAQ and sensor integration, data analysis in Python and MATLAB, and SolidWorks fixture design. Currently inspecting hydraulic components on the production floor at Danfoss Ames; patent holder in sensor-driven agricultural robotics.

EXPERIENCE

Quality Technician - Danfoss Power Solutions (Contract via Aerotek)

Aug 2026 - Present

Aerotek (employer of record), Ames, IA

- Inspect hydraulic components and assemblies at end of line - dimensional, visual, and functional checks against engineering drawings and GD&T specifications using calipers, micrometers, height gauges, and gauge fixtures.
- Identify and document out-of-spec conditions and test discrepancies; escalate to production and engineering, and support containment, root-cause analysis, and corrective action (CAPA/8D).
- Work daily alongside manufacturing, quality, and maintenance teams, building working familiarity with Danfoss hydraulic product families, end-of-line test equipment, and ISO 9001 quality processes.

Analyst - Soil Machine Dynamics Laboratory, Iowa State University

Jul 2026 - Aug 2026

Industry partners • Ames, IA

- Design and run measurement protocols for wear analysis of tillage components, combining physical measurement with image-based analysis (Python, OpenCV, ImageJ); deliver curve-fit trend analyses and comparison reports to industry stakeholders.

Graduate Research Assistant - Iowa State University × Industry Partners

May 2026

Soil Machine Dynamics Laboratory, Ames, IA

- Specified, designed, and commissioned custom test rigs for material flow characterization - modeled fixtures in SolidWorks, fabricated them in-house, and instrumented them with load/force sensors, laser distance sensors, and DAQ hardware for repeatable measurement.
- Created and executed test plans validating simulation predictions against physical experiments (angle-of-repose, hopper discharge, impact plate); quantified agreement using percentage relative error, confidence intervals, and repeatability analysis, and documented outcomes in formal engineering test reports.
- Troubleshoot rig, sensor, and measurement-chain issues to isolate root causes of inconsistent test results; iterated on fixture design and procedure to improve accuracy, repeatability, and operator safety.
- Built flow simulation models (DEM, AcuSolve, HyperMesh) and automated data processing and reporting pipelines in Python, MATLAB, and JMP, improving reproducibility of test data.

EDUCATION

M.S., Agricultural & Biosystems Engineering (Thesis-based)

May 2026

Iowa State University, Ames, IA

Thesis: "Optimization of Dry Fertilizer Spreaders: DEM-Based Material Flow Analysis and Experimental Validation."

B.Tech., Agricultural Engineering

2023

Vignan's Foundation for Science, Technology and Research, Guntur, India

Gold Medal - Best Outgoing Student of the Year, Vignan University.

PATENT

Mechatronic Robotic Device for Self-Regulating and Precision Irrigation

Patent No. 202141009919 • 2021

Designed a ROS-based agricultural robotic system integrating CNN-based weed detection, targeted herbicide application, and closed-loop sensor-driven irrigation control - spanning sensor selection, electronics, and control logic.

TECHNICAL SKILLS

Test & Validation: Test rig / fixture design · test plan creation & execution · design verification & validation · engineering test reports · repeatability & measurement-system analysis · troubleshooting & root-cause analysis (CAPA/8D)

Instrumentation & DAQ: Data acquisition · load/force sensors · laser distance sensors · signal capture & conditioning fundamentals · electronics & soldering · sensor integration (ROS)

Fluid & Flow Analysis: CFD fundamentals · AcuSolve · HyperMesh · DEM modeling · particle & granular flow simulation · engineering calculations

Programming & Data: Python · MATLAB · JMP · Excel · test-data automation · statistical & trend analysis · data visualization · machine vision (OpenCV, ImageJ) · machine learning

CAD & Fabrication: SolidWorks · Fusion 360 · Creo · AutoCAD · prototyping · welding & heat work · CNC milling · lathe · 3D printing & scanning · laser cutting

Quality: Dimensional & functional inspection · GD&T · engineering drawings · non-conformance documentation · ISO 9001

LEADERSHIP, PRESENTATIONS & AFFILIATIONS

Leadership (Iowa State University): Chair, ISTVS Student Chapter; Organizer, Soil Compaction Day; Farm Progress Show and Engineering Career Services Ambassador; World Food Prize Ambassador, Des Moines (2024).

Presentation & Awards: "Developing Digital Twins of Cleaning Shoe Separation of Corn Biomass Particles," CoMFRE Conference, ISU, 2023 · Gold Medal – Best Outgoing Student, Vignan University, 2023 · Memberships: ASABE, ISTVS, CoMFRE